

Quiz 1 - Answer Key

BCY-3A

Total Marks: 15

Question 1: Write the instructions (4 marks)

a. Copy contents of memory location with offset 0025 into AX. [1]

MOV AX, [0025h]

b. Copy AX into memory location with offset 0FFF. [1]

MOV [0FFFh], AX

c. Move contents of memory location 0010 to memory location 002F. [2]

MOV AX, [0010h]

MOV [002Fh], AX

Grading note:

For missing square brackets -0.5 in each step. (direct memory-to-memory moves aren't allowed in 8086)

For any syntax issue -0.5 in each step.

Question 2: Loop program (5 marks)

Write a program to calculate the square of 20 by using a loop that adds 20 to the accumulator 20 times.

Grading note: correct result is ax = 400. marks are split across: accumulator initialization (ax = 0), correct loop counter setup, the repeated add, correct use of loop (or an equivalent structure like dec/cmp/jnz), and reaching the final correct value.

Since people solved this in different ways, grading will vary case by case based on which of these parts are actually present and working, not a fixed rubric applied identically to every submission, but the same scheme is followed for common errors.

The following are some correct solutions:

MOV CX, 20 MOV AX, 0 L1: ADD AX, 20 LOOP L1	MOV AX,0 MOV CX,20 square: ADD AX,20 LOOP square	MOV AX,20 MOV CX,1 line: ADD AX,20 ADD CX,1 CMP CX,20 JNE line
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Question 3: Flags (6 marks)

Initial values: AX = 0x1254, BX = 0x0FFF. Registers carry over cumulatively between parts, each instruction uses whatever value is left from the previous part, not the original values. Only BX in part (a) and AX in part (c) are unaffected, since those registers aren't written to beforehand.

	Result	ZF	CF	SF	OF
a. add ax, 0xEDAB	AX = 0xFFFF	0	0	1	0
b. add ax, bx	AX = 0x0FFE	0	1	0	0
c. add bx, 0xF001	BX = 0x0000	1	1	0	0

Grading note:

2 marks each part: 0.5 per correct flag